

Madhur Lakshmanan

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EXPERIENCE

Causify AI | *Machine Learning Engineer Intern*

July 2025 - September 2025

- Built core components for an LLM-orchestrated AutoEDA platform using Ollama models and FastAPI, implementing JSON/YAML schema profiling, structured outputs, and agent-tool interfaces, which enabled users to analyze datasets through natural-language queries
- Designed modular statistical-analysis tools in scikit-learn for trend, changepoint, outlier, and association detection, translating research methods into executable interfaces for the AutoEDA workflow, which expanded the platform's analytical capabilities
- Built an internal documentation assistant with Dify, automating Markdown ingestion and knowledge-base synchronization via GitHub Actions, which kept documentation up-to-date automatically
- Extended the automated testing framework for parallel and containerized execution by adding pytest multiprocessing hooks and Docker support, which reduced test run time and supported distributed development environments

Thomson Reuters | *AI Technology Intern*

March 2024 - August 2024

- Re-engineered a legacy OmniMark data-processing workflow into production Python and Bash pipelines with parallelization, integrating REST APIs and Microsoft SQL Server and reducing processing time by nearly 70%.
- Developed and evaluated time-series forecasting pipelines in AWS SageMaker, improving forecast accuracy by 18% over the production baseline.
- Designed REST API contracts using OpenAPI and authored technical documentation in Confluence, enabling engineering and legal-product teams to adopt the new services with reduced onboarding time

Mareana (Startup) | *Data Intern*

June 2022 - July 2022

- Authored REST API documentation for 100+ data endpoints, reducing developer onboarding time by 40% and supporting downstream analytics and ML pipeline integration.

PROJECTS

Serverless Research Intelligence Platform

- Python, AWS CDK, Lambda, API Gateway, Aurora PostgreSQL (pgvector), Bedrock, ArXiv API
- Architected a fully serverless research-intelligence platform on AWS that ingests arXiv papers and indexes Amazon Titan embeddings in Aurora PostgreSQL/pgvector for semantic retrieval.
- Designed event-driven services with Lambda and API Gateway that separate ingestion, document processing, retrieval, and conversational inference into independently deployable components.
- Implemented citation-grounded generation returning retrieved passages, paper metadata, and similarity scores alongside answers for source-traceable outputs.
- Provisioned complete infrastructure via AWS CDK, including IAM policies, Secrets Manager, deduplication, structured logging, and automated deployment.

Financial and Regulatory Document Intelligence Platform

- Python, OpenAI GPT, LangChain, SEC EDGAR API, Tableau
- Built an SEC document-intelligence pipeline covering 15+ years of 10-K/10-Q filings across 500+ public companies, handling ingestion, preprocessing, extraction, and retrieval.
- Designed schema-constrained GPT-4 extraction pipelines using few-shot prompting and validated JSON schemas to identify risk factors, litigation disclosures, financial metrics, and material events.
- Orchestrated multi-stage LLM workflows for extraction, validation, retrieval, summarization, and report generation, producing auditable structured outputs.

EDUCATION

University of Maryland

May 2026

Master of Science, Applied Machine Learning (GPA: 3.8/4.0)

Vellore Institute of Technology

May 2024

Bachelor of Technology, Electronics and Communication Engineering (GPA: 3.2/4.0)

SKILLS

Languages: Python, SQL, Java, C/C++, Bash

ML / GenAI: PyTorch, scikit-learn, Hugging Face, LangGraph, LangChain, RAG, pgvector, vLLM

Cloud / Infrastructure: AWS (SageMaker, Lambda, Bedrock, CDK, API Gateway, S3), Docker, Kubernetes, GitHub Actions, Airflow

Backend / Data: FastAPI, PostgreSQL, PySpark, OpenSearch, REST APIs, Microsoft SQL Server